

PREDICTIVE AND TACTICAL INTELLIGENCE SYSTEM FOR ATHLETE PERFORMANCE USING VISION-BASED TRACKING AND WEARABLE DATA ANALYTICS

Capstone Project Proposal

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Mentor Consent Form

I hereby agree to be the mentor of the following Capstone Project Team

Project Title: Predictive and Tactical Intelligence System for Athlete Performance Using Vision-based Tracking and Wearable Data Analytics

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1 Project Overview

1.1 Background

Sports is a quick evolving domain where modern technology can be leveraged to improve athlete performance, refine training methodologies, minimize injury risks, and support data-driven decision-making. Recent advancements in AI/ML have transformed the landscape of sports analytics. Many existing sports analytics tools focus on isolated data sources or provide only post-game analysis rather than real-time performance monitoring. Therefore, there is a growing need for intelligent systems that can integrate multi-modal data and provide real-time predictive and tactical insights.

1.2 Introduction to Problem Statement

To address this gap, a comprehensive sports analytics framework is conceptualized. This predictive and tactical intelligence system is designed primarily for football, though its architecture can be adapted to other similar sports. The system aims to monitor, analyze, and enhance athlete performance in real time by leveraging vision-based tracking and wearable data.

The primary functionality of this tool is the fusion of multiple data streams. Computer vision methods are used to monitor player motion and tactical positioning across the field, while wearable devices capture key physiological and spatial metrics. By integrating these visual and sensor-based data sources, machine learning algorithms synthesize insights, shifting performance evaluation from manual observation toward a predictive analytical framework that supports coaches and improves decision-making.

2 Need Analysis

In the competitive world of football, figuring out exactly how players are performing and how well teams are executing their tactics is a difficult task. For years, we have relied mostly on coaches' eyes and post-match video review. But as the game has gotten faster, smarter, and more physically demanding, those methods aren't sufficient anymore. The naked eye cannot catch and coordinate the various micro-movements at a blink. There is a huge gap between what athletes are doing and the kind of data driven decisions that could enhance their performance. That is exactly why we need a system that gives coaches real-time insights instead of just guesses.

Injury prevention is another massive issue. Non-contact injuries like hamstring strains causes players to miss matches and medical bills. These injuries don't just happen out of nowhere, they are usually preceded by building fatigue that no one notices in the moment. Most tools don't combine internal data from wearable sensors with external movement data from cameras. Therefore, we need a single, smart platform that integrates everything together in real time and actually warns us before player gets hurt.

Tactically, coaches only see what is happening from the sidelines and few angles. Even with all the expected goals and possession prediction models out there now, we still don't have a proper live system that tracks every player continuously and feeds that into instant tactical advice. That is where this project emerges. We are building a predictive intelligence system that fuses computer vision with wearable sensor data using advanced machine learning.

3 Literature Review

The integration of computer vision, wearable sensor data, and machine learning techniques has been widely explored in the domain of sports analytics, particularly for football performance monitoring and injury risk assessment. Existing systems and research efforts have primarily focused on either vision based tracking or wearable load analytics in isolation, with limited attempts at multimodal fusion for real-time predictive and tactical intelligence.

3.1 Vision-Based Tracking in Football

Recent advances in computer vision have provided various techniques for tracking players. Researchers have relied mainly on object detection models, particularly YOLO (You Only Look Once) [1] architecture, combined with tracking algorithms like ByteTrack [2]. For instance, Yousef et al. [3] demonstrated a cost-effective CV system specially for sport academies, achieving a 96% accuracy rate in player tracking. However, their system had a common setback in purely optical approaches i.e. poor ball detection rates (36%) due to occlusion and low resolution video. Zheng et al. [4] reviewed these challenges, noting that although Transformer + LSTM hybrids can effectively extract non contact metrics like speed and pass prediction, small object detection and occlusion remain significant setback in uncontrolled match environments.

3.2 Wearable Technology and Injury Prediction

While computer vision provides extensive external data, wearable technology remains the gold standard for monitoring athlete workload and predicting injury risks. Recent studies have shown machine learning being applied intensively to data derived from Global Positioning System (GPS) units and micro-electromechanical systems (MEMS). Pilka et al. [5] successfully utilized GPS data in professional football following an XGBoost algorithm approach to acute:chronic workload ratios (ACWR) and high-speed running (HSR) metrics. Similarly, Leckey et al. [6] demonstrated that tree based models (Random Forest and XGBoost) are highly effective at evaluating external load data to assess injury risk. However, both studies notice a significant limitation that relying solely on wearable

data ignores spatial and kinematic data and lacks the broader positional context that vision based systems provide.

3.3 Tactical Dynamics and Multimodal Fusion

To bridge the gap between individual player metrics and team wide strategies, researchers have explored AI models capable of analyzing collective team dynamics. Teixeira et al. [7] leveraged CNN and LSTM architectures to analyze space control and formations from tracking data, achieving up to 90% accuracy in strategy optimization. Towards multimodal integration, Honda et al. [8] proposed a system for pass receiver prediction that fuses video footage with player trajectories. By combining a 3D CNN for video body motion with an LSTM for trajectories and a Transformer model, their multi architecture approach yielded a 13% accuracy improvement over single-modality models.

Table 1: Summary of Literature Review

References	Concepts Highlighted	Technique(s) used	Significance	Limitations
[4]	Player/ball detection, motion prediction, tactical analysis in football videos	YOLOv8 + ByteTrack; Transformer-LSTM hybrids	Non-contact real-time metrics (speed, possession, pass prediction); scalable alternative to wearables	Occlusion, low resolution, small-object detection issues; limited annotated datasets
[3]	CV system for academies: player tracking, possession, speed metrics	YOLOv8 detection + ByteTrack; colour-based team assignment	Affordable for modest clubs; high player detection accuracy (96%)	Poor ball detection (36%); no wearable integration or predictive modules

[5]	Injury prediction from GPS wearables in professional football	Catapult GPS data; XGBoost on load ratios (ACWR, HSR)	F1-score 94.4%; enables microcycle workload adjustment	External load data only; ignores vision-derived positional context and internal signals
[6]	Machine-learning approaches to sports injury risk (football dominant)	Tree-based models (XGBoost, Random Forest); GPS/wearable external load	Best performance with multimodal data (AUC 0.77); identifies current trends and gaps	Small heterogeneous datasets; limited real-time deployment and clinical validation
[7]	AI for tactical behaviour and collective team dynamics	CNN/LSTM, graph metrics (centrality), clustering/XGBoost on tracking data	Formation and space-control analysis (up to 90% accuracy); supports strategy optimisation	Mostly post-match analysis; lacks full multimodal real-time predictive integration
[8]	Multimodal pass-receiver prediction fusing video and trajectories	3D-CNN (video body motion) + LSTM (trajectories) + Transformer	+13% accuracy improvement from vision features; provides tactical insight	Short-term prediction only; no injury-risk forecasting or long-term performance trends

4 Research Gaps

Some of the key research gaps identified from the literature review are as follows:

- Most existing systems and research address player tracking and performance analysis in isolation, either using computer vision techniques or wearable sensor data alone. Very limited work exists on effective multimodal fusion.
- Commercial platforms such as Genius Sports (optical tracking) and Catapult Sports (wearable GNSS/IMU systems) deliver robust individual components but remain highly expensive, proprietary, and inaccessible to resource constrained teams and university clubs.
- The majority of tools provide insights only in post match mode. Near-real-time tactical decision support modules that generates recommendations during training remain underdeveloped.

5 Problem Formulation

In professional football, improving player performance, team tactics and preventing non contact injuries are important goals that are currently hindered by isolated analysis techniques. Traditional evaluation methods heavily rely on subjective human observation or post-match video analysis, which fail to capture the complex, high speed spatial dynamics and subtle biomechanical variations occurring during live competition. Also, strategic decisions made on the pitch are often reactive rather than proactive.

Furthermore, an important technological gap exists in the real time integration of isolated data streams. While optical tracking is utilized to map external kinematic data (player positioning, velocity, and acceleration) and wearable sensors are deployed to monitor internal physiological workloads (such as heart rate and SpO2 levels), these systems traditionally operate in isolated modalities. Not being able to synchronize and fuse these multimodal data during active gameplay prevents coaching and medical staff from obtaining a holistic understanding of an athlete’s immediate physical state.

Because internal exertion is not visible in external movements in real time, the early indicators of physiological fatigue are frequently missed. This gap directly leads to a high occurrence of preventable, non-contact injuries. Also, the lack of immediate spatio-temporal data restricts the ability of coaches to make data-driven, tactical adjustments to exploit opponent vulnerabilities.

Therefore, the central problem addressed by this project is the lack of a unified intelligence platform capable of continuously tracking athlete movements via computer vision while simultaneously analyzing physiological sensor data. A robust system must be engineered to analyze these multimodal inputs using machine learning algorithms. By resolving this integration problem, insights regarding injury risk and optimal tactical positioning can be delivered, thereby transforming isolated sports data into comprehensive, predictive intelligence.

6 Objectives

The following objectives have been defined for the successful completion of the capstone project:

- To develop an integrated data processing framework that combines vision based player tracking with wearable sensor data streams for athlete monitoring in football.
- To use a wearable hardware to extract physiological and spatial data from multiple sensors like GPS, accelerometer, pulse oximeters, etc.
- To design and implement machine learning models capable of analyzing the fusion of multi-modal data to predict performance trends, player and team wise and a tactical decision module that generates recommendations based on the performance.
- To evaluate the developed system on available datasets or controlled test scenarios.

7 Methodology

The methodology adopted to achieve the four defined objectives follows a modular pipeline approach. The system architecture processes vision-based tracking and wearable sensor data streams and applies machine learning for prediction and tactical support. The overall workflow is illustrated in Figure 1.

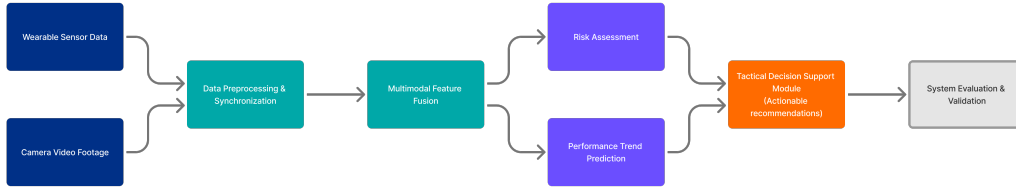


Figure 1: Block Diagram of the Proposed Predictive and Tactical Intelligence System

The stepwise methodology is described as follows:

- 1. Development of the integrated data processing framework (Objective 1):** Video footage is captured and processed using object detection (YOLOv8) followed by multi-object tracking (ByteTrack) to extract real-time player positions, speeds, trajectories, and ball movement. Simultaneously, wearable sensor data (GPS, accelerometer, pulse oximeter) is consumed and synchronized. Raw data streams are cleaned and stored in a unified pipeline.
- 2. Construction of the wearable hardware prototype (Objective 2):** A custom wearable device is built using a microcontroller integrated with GPS module and sensors like accelerometer, and pulse oximeter. The hardware collects physiological parameters (heart rate, SpO2) and spatial data (position, acceleration) at regular intervals and then the data is fed into the processing framework.
- 3. Design and implementation of machine learning models and tactical decision support module (Objective 3):** Features extracted from vision tracking (positional metrics, possession zones, speed) and wearable sensors (distance covered, high-speed running, player load) are combined. Various supervised models for

regression/classification and time-series trends are trained on the fused dataset to predict player-wise and team-wise performance trends. Outputs from the prediction models are fed into a decision engine. Rule based logic along with clustering or simple classifiers generates recommendations, such as adjustments, substitutions or redistribution based on analysed performance.

4. **Evaluation of the developed system (Objective 4):** The complete pipeline will be tested on publicly available football tracking datasets (e.g. SoccerNet) and controlled test scenarios recorded during local training sessions.

This modular design allows seamless development and testing of each component. The approach remains flexible to subtle changes as required during project development timeline.

8 Datasets

Table 2: Datasets

Name	Features	Size	Year	Organization
SoccerNet [9]	Raw footages of football matches + annotations of players, ball and field.	17.02GB	2018	University of Liège and KAIST

9 Work Plan

The work plan for the successful execution of the capstone project has been scheduled from February 2026 to December 2026. The Gantt chart representation below shows the major activities along with their planned timelines, considering academic constraints such as examinations and semester breaks.

Table 3: Project Work Plan (Gantt Chart - February to December 2026)

Sr.	Activity	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	Project Identification, Planning, Literature Review											
2	Design and Prototyping of Wearable Hardware											
3	Implementation of Vision-based Player Tracking Module											
4	Data Collection, Preprocessing and Feature Fusion Pipeline											
5	Development and Training of Machine Learning Models											
6	Development of Tactical Decision Support Module											
7	System Integration, Real-time Testing and Debugging											
8	Final Optimization and Documentation											

10 Project Outcomes & Individual Roles

The capstone project will deliver the following outcomes upon successful completion:

- A fully integrated data processing framework that combines vision based player tracking with wearable sensor streams for real-time athlete monitoring in football.
- A wearable hardware prototype capable of collecting physiological (heart rate, SpO2) and spatial (GPS, acceleration) data.
- Trained machine learning models for predicting player-wise and team-wise performance trends and a tactical decision module that generates recommendations such as formation adjustments, substitutions, and redistribution.
- Comprehensive system evaluation results on public datasets and controlled test scenarios, along with a simple visualisation dashboard/app.

The individual roles and responsibilities of the team members are as follows to ensure balanced contribution:

Table 4: Team Member Roles and Responsibilities

Sr.	Name	Role and Responsibilities
1	Siddartha Nepal	Overall project coordination and development of the vision-based tracking module and tactical engine.
2	Raghav Agrawal	Development and testing of the wearable hardware prototype, including data acquisition and synchronisation.
3	Bibek Kumar Thagunna	Implementation of machine learning models using multimodal data sources.
4	Suyogya Sedhai	Development of the tactical engine and visualisation dashboard.
5	Om Narayan Yadav	System evaluation, documentation, and execution of testing scenarios.

11 Course Subjects

The successful execution of this capstone project will utilise conceptual knowledge and practical skill acquired from the following courses and topics:

- **Machine Learning:** Concepts of supervised learning, regression, classification, ensemble methods, and hyperparameter tuning applied to build ML models for player-wise and team-wise performance trend prediction.
- **Computer Vision:** Techniques for object detection and tracking used to implement YOLOv8 for player and ball movement extraction.
- **Deep Learning:** Knowledge of recurrent neural networks and time-series modelling applied to LSTM networks for analysing sequential multimodal data streams.
- **Engineering Design Project:** Practical skills in hardware prototyping, sensor interfacing and microcontroller programming for building the wearable hardware prototype.
- **Artificial Intelligence:** Basic concepts of decision making systems, rule-based engines, and clustering to develop the tactical decision module.
- **Software Engineering:** Principles of modular system design, SDLC, integration testing, documentation, and project management applied throughout the development phases.

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